

Does Adding More Features Necessarily Improve Prediction?

An investigation of informative, irrelevant and redundant features in house-price prediction (Team 04 – Feature Selection)

1. Team Members

Pranav S Nair	Ananthakrishnan K V	Anna Mariya Shibu	Aishwarya Pramod Nair
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2. Research Question

Assigned question: Does adding more features necessarily improve prediction?

Operational question: How does the type of feature added – informative, irrelevant or redundant – affect test error, generalization gap and stability of a regression model, and does the effect depend on the learning algorithm (a regularized linear model versus a tree ensemble)?

3. Hypotheses

- **H1:** Adding informative features improves predictive performance, because they supply new signal about the target.
- **H2:** Adding irrelevant features does not systematically improve performance and may increase error or variability, because a model can fit chance patterns in them.
- **H3:** Adding redundant features gives little or no benefit, because their information is already present.
- **H4:** The effect of feature addition differs between Ridge Regression and Random Forest, because they use features in fundamentally different ways.

4. Theoretical Background

Bias-variance trade-off. Expected test error decomposes into bias², variance and irreducible noise. Informative features reduce bias by letting the model represent more of the true signal. Every extra input, however, also adds something to be estimated from finite data, which tends to raise variance (Hastie et al., 2009). More features therefore help only if the bias reduction outweighs the variance cost.

Irrelevant and redundant features. An irrelevant feature carries no information about the target; any split or weight assigned to it can only reflect sampling noise. A redundant feature is highly correlated with an existing one, so it adds no new information but creates multicollinearity (Guyon & Elisseeff, 2003).

Ridge regression minimizes squared error plus $\alpha \cdot \sum w_j^2$ (Hoerl & Kennard, 1970). The L2 penalty shrinks coefficients, which limits the variance a useless feature can introduce, and it spreads weight across correlated features instead of letting them compete. Ridge is therefore expected to be comparatively tolerant of both feature types.

Random Forest averages many deep, unpruned trees grown on bootstrap samples (Breiman, 2001). It has no coefficient-shrinkage mechanism. In deep nodes with few samples, a noise feature can offer a split that looks good on the training data by chance, and such spurious splits do not transfer to new data. Redundant copies, by contrast, are interchangeable and mainly share importance with the original. Trees are also deep enough to fit the training data very closely, so a large train-test gap is expected regardless of features.

5. Dataset

Dataset. The Ames Housing dataset (De Cock, 2011): 2,930 residential sales in Ames, Iowa (2006–2010) described by 82 columns.

Target: SalePrice (USD). **Key features:** Overall Qual (1–10 quality rating), Gr Liv Area, Total Bsmt SF, Garage Cars/Area, Year Built, Year Remod/Add and Neighborhood, among 81 predictors (numerical and categorical).

Why it suits this investigation. (i) It has enough rows (2,344 for training) to support twelve model-condition runs without severe sample-size effects; (ii) it contains a genuine mix of strongly, weakly and barely informative real features, so "informative" can be defined from domain knowledge; and (iii) it lets us inject synthetic features whose relevance is known by construction (pure noise, or near-copies of real columns), giving a controlled ground truth that real data rarely offers.

Limitations and biases. All sales come from one city and four years, so conclusions may not transfer to other markets. SalePrice is right-skewed. Many NaN values encode absence (e.g., no pool, no alley) rather than missing data, and are handled by imputation.

6. Experimental Design

Changed: only the feature set (six conditions, Table 1). **Measured:** Test RMSE (primary metric), Test MAE, Test R², Train RMSE and the generalization gap (Test RMSE – Train RMSE). **Controlled:** one fixed 80/20 split (random_state = 42; 2,344 training and 586 test houses) reused by every condition; identical hyperparameters (Ridge $\alpha = 10$; Random Forest with 300 trees, random_state =

42); and all preprocessing inside scikit-learn Pipelines fitted on training data only, which prevents leakage (median imputation for numeric columns; most-frequent imputation and one-hot encoding for categorical columns; standardization for Ridge only, since trees are scale-invariant). **Compared:** A→B isolates informative features; C→D, C→E and C→F isolate irrelevant, redundant and combined additions.

The 10 irrelevant features are independent $N(0,1)$ draws. The 5 redundant features are scaled copies of existing columns plus small noise (e.g., $1.02 \times \text{Gr Liv Area} + N(0,10^2)$); their Pearson correlations with the originals were verified as 0.990–1.000.

Table 1. Experimental conditions

ID	Feature set	Raw	After encoding	Content
A	Core informative	10	10	Hand-picked strong numeric predictors (baseline)
B	Informative-20	20	20	A + 10 further genuine numeric features
C	All original	81	302	Every original predictor (numeric + categorical)
D	C + irrelevant	91	312	C + 10 random noise features
E	C + redundant	86	307	C + 5 near-duplicate features ($r \geq 0.99$)
F	C + both	96	317	C + 10 irrelevant + 5 redundant

7. Results

Table 2. Test performance of all twelve model-condition runs (single split; RMSE and gap in USD)

ID	Raw	Ridge Test RMSE	Ridge R^2	Ridge gap	RF Test RMSE	RF R^2	RF gap
A	10	39,549	0.805	4,260	31,658	0.875	20,877
B	20	36,621	0.833	3,154	27,758	0.904	17,731
C	81	29,161	0.894	6,210	26,711	0.911	16,865
D	91	29,082	0.895	6,177	27,560	0.905	17,504
E	86	29,176	0.894	6,221	26,893	0.910	16,974
F	96	29,087	0.894	6,182	27,632	0.905	17,437

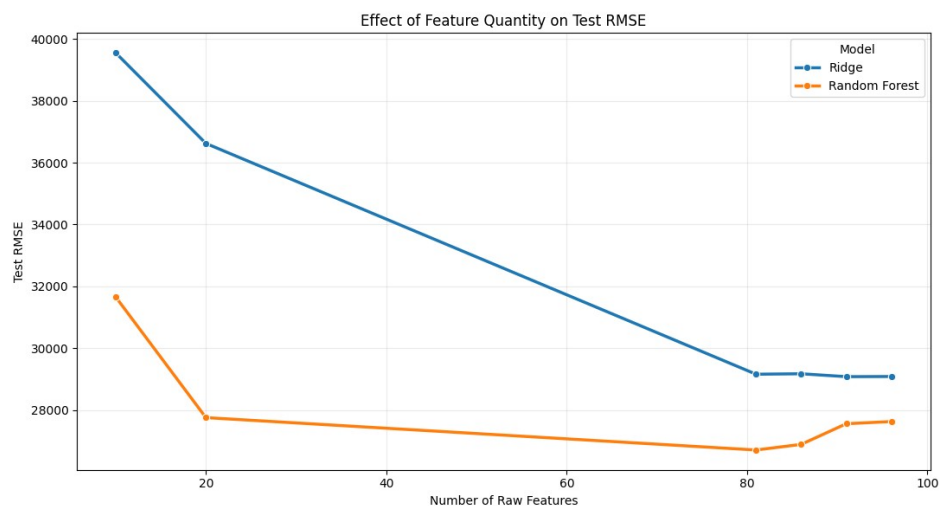


Figure 1. Test RMSE versus number of raw features (A→B→C add informative features; C→E→D→F add non-informative ones).

Table 3. Change in Test RMSE relative to the reference condition

Comparison	Feature type added	Ridge Δ RMSE	Random Forest Δ RMSE
A → B	+10 informative	−2,928 (−7.4%)	−3,900 (−12.3%)
C → D	+10 irrelevant	−78 (−0.3%)	+849 (+3.2%)
C → E	+5 redundant	+15 (+0.05%)	+181 (+0.7%)
C → F	+10 irrelevant, +5 redundant	−74 (−0.3%)	+920 (+3.4%)

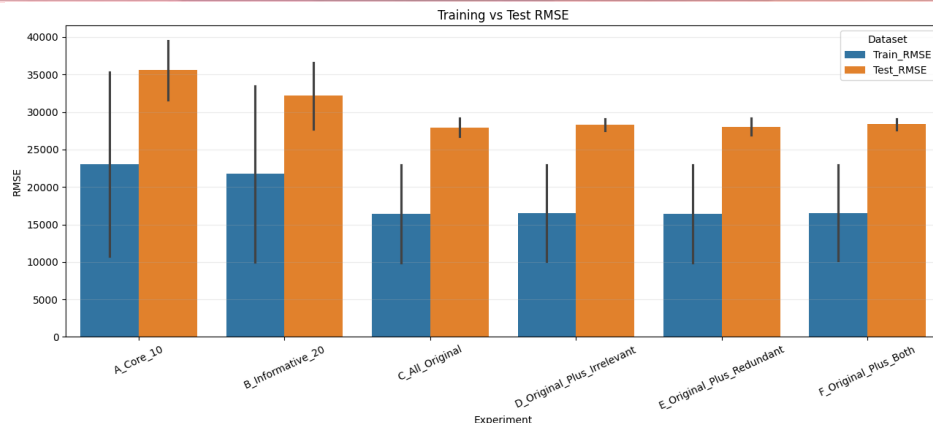


Figure 2. Training versus test RMSE for every condition and model (the gap is the visible overfitting signal).

8. Analysis and Interpretation

H1 – informative features help (supported). Moving from 10 to 20 informative features cut Test RMSE by 7.4% (Ridge) and 12.3% (Random Forest), and the generalization gap also fell for both models (–1,106 and –3,146). The gain was therefore real generalization rather than better memorization. Adding all 81 original features helped further (Ridge –26% and Random Forest –16% versus A), consistent with the bias-reduction argument: the 10-feature Ridge model has a training RMSE of 35,289, close to its test RMSE, which is the signature of an underfit, high-bias model with too little signal to work with.

H2 – irrelevant features (supported for Random Forest; neutral for Ridge). Ten pure-noise columns changed Ridge by only –78 (–0.3%), which is too small to interpret as a benefit, but raised Random Forest error by 849 (+3.2%) and its gap by 638. This matches the theory: the L2 penalty keeps Ridge weights on noise columns close to zero, whereas unpruned trees can split on noise in small nodes, fitting chance patterns that do not generalize. Because the default Random Forest regressor considers all features at each split, the mechanism is spurious-split selection in deep nodes, not random feature subsampling. We did not inspect feature importances, so this remains a theoretically motivated explanation rather than a directly verified one.

H3 – redundant features (supported). Five near-duplicates changed test error by only +15 (Ridge) and +181 (Random Forest). They contain no new information ($r \geq 0.99$), so at most they redistribute weight (Ridge) or importance (trees) between original and copy. The small Random Forest increase is comparable to the size of differences that could arise from resampling noise, so we do not treat it as a real effect.

H4 – model dependence (supported, with caveats). The two models responded in opposite directions to noise (–78 versus +849). Random Forest also showed a generalization gap of about 17,000 in every condition, roughly three times Ridge's, even with only 10 features. This gap is a property of a flexible, high-variance learner rather than of any added feature, and it shows why the train–test gap must be read together with test error. Note that Ridge's gap rose from B (3,154) to C (6,210) while its test error fell: 302 parameters fit training data more tightly, but the extra genuine signal outweighed the added variance.

Why the plateau at C? Once every genuinely informative column is present, further columns can only add variance or nothing. The curves in Figure 1 therefore fall steeply while informative features are added and then flatten or drift upward, which is the bias–variance trade-off made visible.

9. Critical Evaluation and Limitations

- **Single train–test split and single seeds.** All results come from one split, one Random Forest seed and one draw of noise features. Large effects (A→B, about 3,000–3,900) are unlikely to reverse, but small ones (Ridge C→D –78; Random Forest C→E +181) lie within plausible noise, and even the +849 Random Forest effect is suggestive rather than established. Repeated splits or cross-validation would quantify this.
- **Synthetic feature types.** Independent Gaussian noise is a best-case "clean" irrelevant feature; real uninformative features are often weakly correlated with the target. Redundancy was built by linear scaling plus noise, so non-linear redundancy is not covered.
- **Condition C is not purely informative.** The 61 extra original columns vary from very useful to nearly useless, and one-hot encoding raises the model input from 81 to 302 columns. The A→B→C effect therefore mixes information gain with a change in dimensionality and feature type. Irrelevant and redundant features were also tested only on top of an already saturated baseline; their effect might be larger starting from A.
- **Identifier columns were retained.** The dataset's Order and PID columns were not removed before modelling (the notebook's Id-drop step targets a column that does not exist in this file), so they enter conditions C–F as numeric features. Because PID is assigned geographically, it may act as a weak proxy for location. It affects all C–F conditions equally, but absolute errors may be slightly optimistic, and the run should be repeated without them.

- **Limited scope.** Two algorithms, fixed hyperparameters (no tuning), one city and 10–15 added features against 2,344 training rows. The harm from noise features is expected to grow when the sample is small relative to the number of features; this was not tested.

10. Conclusion

What happened. No – adding more features does not necessarily improve prediction. Informative features improved both models substantially (A→B: –7.4% and –12.3% Test RMSE). Adding 10 irrelevant features left Ridge unchanged but raised Random Forest error by about 3%, and redundant features made no meaningful difference to either model.

Why. Improvement comes from new information, not from feature count: informative features reduce bias, whereas irrelevant features add only variance and redundant ones add nothing new. Ridge's L2 shrinkage neutralizes useless inputs; unpruned trees have no such mechanism and can fit noise in deep splits.

Confidence. High that informative features help and that redundant features do not matter here; moderate that irrelevant features hurt Random Forest more than Ridge (the direction agrees with theory, but the evidence rests on one split and one seed). Cross-validation with repeated noise draws, feature-importance inspection and removal of the identifier columns would raise confidence further.

11. AI Usage Declaration

AI tool(s) used	Claude (Anthropic), ChatGPT (OpenAI), and Kiro.
Purpose of using AI	Code structure, debugging and explanation; documentation; drafting and formatting the report text from the team's notebook results.
Parts assisted by AI	Notebook code and commentary; report wording and layout. Experimental design decisions and interpretation are the team's responsibility.
How output was verified	All notebook cells were executed by the team; every number was cross-checked against notebook outputs; theory claims were checked against standard references.
One suggestion modified, rejected or corrected	An AI-generated explanation initially attributed Random Forest's sensitivity to irrelevant features to random feature subsampling at each split. This was corrected because scikit-learn's default Random Forest Regressor considers all features when determining each split. The explanation was revised to describe the effect as possible spurious split selection on noise features in deeper tree nodes.

12. References

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